

Cole Matthews

Durham, NC | Cole.Matthews@duke.edu | colematthe.github.io
British and Canadian citizen | OPT-eligible for summer internships, no sponsorship needed

EDUCATION

Duke University

B.S.E. Mechanical Engineering, Aerospace Engineering Certificate

Durham, NC

Expected May 2028

- **Coursework:** Thermodynamics, Fluid Mechanics, Dynamics, Mechatronics, Differential Equations; Controls (Spring 2027).
- **Labs:** engine cycle analysis (Brayton, dual cycle), spring-pendulum dynamics, op-amp filter design, DAQ instrumentation.
- **Study abroad:** UNSW Sydney, Aug–Dec 2026.

City of London School

Predicted A-Levels: Mathematics (A*), Further Mathematics (A), Physics (A*), Economics (A*)

London, UK

Sep 2019 – Jul 2024

- **GCSEs:** 10 at grade 9/A*; Lord Mayor's Prize for GCSE results (2022).

EXPERIENCE

ADAG Group

CFD & Design Intern

Montreal, Canada

Jun 2026 – Jul 2026

- Learned Cadence Reality DC, a data-center computational fluid dynamics (CFD) tool, from scratch, then built and solved full thermal and airflow models for a **160 MW AI data-center campus**.
- Ran a three-tier CFD fidelity study of a **600 kW** fan wall cooling unit, setting $\pm 5\%$ tolerances to find the lowest-cost model that reproduced the high-fidelity result.
- Modeled a rear-door heat exchanger for GPU racks at **~ 50 kW/rack**, trading heat rejection against air-side pressure drop.

McLaren Applied

Motorsport Intern, Systems Engineering

Woking, UK

May 2025 – Jul 2025

- Built electronic control unit (ECU) validation workflows for the **TAG-700, the standard ECU on every 2026 Formula 1 car**; calibrated lambda sensors on a Bosch emulator.
- Designed **12+ TAG-700 and PCU-700 test cases** across multiple vehicle programs: fault detection, UI, voltage and timekeeping.
- Helped build an early **Human-in-the-Loop**: a briefcase-sized trackside ECU diagnostics rig (PEAK PCAN, System Monitor).

Duke Motorsports (Formula SAE)

Powertrain Engineer

Durham, NC

Sep 2024 – Present

- Calibrate the engine and configure the drivetrain to improve powertrain performance.
- Integrate powertrain components with the chassis and electrical sub-teams ahead of competition.

BAE Systems Submarines

Engineering Intern

Barrow-in-Furness, UK

Oct 2023

- Shadowed engineering and manufacturing teams building the UK's Dreadnought and Astute-class nuclear submarines.

SELECTED PROJECTS

Engine Misfire Diagnosis from In-Cylinder Pressure Data | Python

Spring 2026

- Processed in-cylinder pressure data from a 3.5 hp Briggs & Stratton engine in Python (peak detection, **crank-angle reconstruction, PV-loop integration**) and produced CAD for its intake manifold adapter.
- Ran three misfire tests that traced the flat PV loop to incomplete combustion; **0 of 12** cycles passed.

Low-Cost Thermoelectric Refrigerator | Engineering design course, Duke

2024 – 2025

- In a team of four, built the insulated box and wired the Peltier cooling system of a low-cost fridge for households in Bangladesh; cooled **1.49 ft³** to **50.2 °F** (target 35 to 40 °F) and traced the shortfall to the Peltier, not the insulation.

Optimizing 3D Printing for Use in Space | CREST Gold Award, British Science Association

2023 – 2024

- Tested **30 3D printer settings** across 7 variables in torsion, bending and impact (90+ hours).
- Built a **Charpy impact tester** from scratch; measured impact energy nearly doubling (3.2 to 6.3 J) with layer orientation.

Simplicity | Founder, student wellbeing platform

2022 – Present

- Built a wellbeing platform used by two London schools; won **£10,000** in the Transform Your Society competition.

TECHNICAL SKILLS & RECOGNITION

Aviation: FAA private pilot training (Cessna 152, Piper PA-28); UK CAA drone license

Engineering: CFD (Cadence Reality DC), thermal modeling, CAD (SolidWorks, Onshape, Fusion 360), FDM 3D printing, materials testing, ECU validation, engine calibration, CAN bus (PEAK PCAN), ATLAS telemetry, data acquisition (DAQ)

Software: Python (signal processing, data analysis), MATLAB

Recognition: CREST Gold Award (2024); Senior British Physics Olympiad Silver Medal (2023);

Ivy House Leadership Development Award (2023)

Memberships: Royal Aeronautical Society (Student Affiliate); Vertical Flight Society

Interests: Aviation, motorsport, space and rocketry, skiing, football (Chelsea FC), sustainability, travel, exploring new cuisines